

Carmen Lucca DiMario

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EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Master of Science in Computer Science

GPA: 4.0

Graduation:

May 2027

Bachelor of Science in Computer Science

GPA: 3.97

May 2026

TECHNICAL SKILLS

Programming Languages: Swift, Java, TypeScript, JavaScript, Python, C, C++, SQL, Bash

Mobile & Front End: SwiftUI, iOS, StoreKit 2, React, React Native, Next.js, REST APIs, JSON

Tools & Practices: Git/GitHub, Agile/Scrum, PostgreSQL, AWS, Auth0/JWT, UNIX

PROFESSIONAL EXPERIENCE

Garmin: Embedded Software Development Intern

May 2026 - August 2026

- Implemented Clang compiler support as a replacement for legacy ARMcc toolchains, modernizing the build for embedded avionics software
- Extended embedded library support to 64-bit architectures, resolving data type alignment and pointer-width issues to broaden platform compatibility
- Developed a system test suite for a proprietary communication interface, validating real-time data transmission and protocol correctness

JRC Integrated Systems: RISE Intern/CO-OP Participant

June 2025 - May 2026

- Rebuilt the Accuro test suite from hard-coded shell scripts into a Python framework with parallel execution, selectable test subsets, and configurable output, cutting a multi-hour run to roughly 25 minutes
- Upgraded the application's PETSc integration across a major version boundary to enable GPU-accelerated solvers for its core computations
- Developed and optimized software tools for radiation-hardened semiconductor testing

ivDash: Full Stack Development Intern

May 2024 - August 2024

- Led development of an administrative dashboard consolidating workflows for doctors, nurses, and product providers into a single system
- Built interactive web interfaces in Next.js against an existing Symfony PHP backend
- Streamlined Auth0 JWT authentication across both the Symfony and Next.js applications

PROJECT EXPERIENCE

PFactor: iOS Application

August 2025 - Present

- Built and shipped an AI oral-exam simulator to the App Store in Swift and SwiftUI, owning product design, backend, and client
- Architected a serverless backend handling Sign in with Apple, StoreKit 2 subscriptions, and graceful offline degradation via automatic local-scoring fallback
- Created an adaptive examiner engine scoring responses 1-5, generating targeted follow-ups on weak areas, and delivering per-category performance debriefs

Engineering Physics Propulsion Laboratory: AIRHOUND [Middleware Lead]

September 2024 - May 2026

- Designed a modular four-node ROS2 perception-to-control pipeline (detection, tracking, PINN prediction, PX4 bridge) communicating with a PX4 FMUv6X flight controller over Micro XRCE-DDS
- Diagnosed DDS image-transport serialization as the pipeline bottleneck and replaced it with direct V4L2 capture, cutting mean detection latency from 234 ms to 58.8 ms and eliminating all dropouts at 1280x720

PUBLICATIONS

DiMario, C., Bacha, R., & Butka, B. "Combating Senior Scams Using a Large Language Model-Created Rubric." ASSE '24, ACM, 2024. <https://doi.org/10.1145/3702138.3702140>

Malarchick, R., DiMario, C., et al. "Predictive Target Pursuit for Autonomous UAVs Using RF-DETR with Depth-Aware State Estimation and Physics-Informed Trajectory Prediction." Proc. SPIE 14030, 140300O (2026). <https://doi.org/10.1117/12.3094732>

EXTRACURRICULAR ACTIVITIES

- **ERAU Division II Cross-Country/Track:** Approximately 30 hours of training weekly
- **FAA Licenses:** Part 107 Remote Pilot Certification

<https://dimario.dev>